Chemistry IA

Research Question:

How does boiling time (0–8 minutes) affect the concentration of ascorbic acid in spinach, as determined by iodine redox titration?

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1 Introduction

Baby spinach (spinacia oleracea) is widely used around the world as a nutrient-dense leafy green, highly valued for its high content of vitamins and minerals.[1] One of its common uses is as a key nutritional ingredient in morning smoothies and one of its most important nutrients is vitamin C, also known as ascorbic acid, a water-soluble vitamin and antioxidant essential for collagen synthesis, immune function, iron absorption, and protection against oxidative stress.[2] As an antioxidant, ascorbic acid helps neutralize free radicals in the body, thereby reducing cellular damage and contributing to overall health. Because humans cannot synthesize ascorbic acid, it must be obtained through dietary sources, making vegetables such as spinach important components of a balanced diet.[2] Ascorbic acid is not only nutritionally important but also chemically sensitive to environmental factors such as heat, oxygen, and water exposure. In everyday cooking, particularly when preparing smoothies, many individuals boil baby spinach for varying durations to reduce the risk of E. coli or other food-borne bacteria. However, the heat and exposure to oxygen and water during this process can negatively impact ascorbic acid content, as it promotes leaching into the surrounding water due to its high water solubility and increases the rate of oxidation for the ascorbic acid. In this oxidative step, ascorbic acid loses two protons and two electrons to become dehydroascorbic acid (DHA).

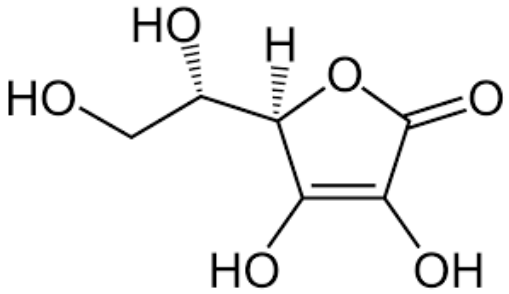


Figure 1: Chemical Diagram of Ascorbic Acid (Vitamin C)

While the body can actually convert DHA back into ascorbic acid, DHA is very unstable and the nutritional loss is a result of the "spent" antioxidant potential. Ascorbic acid functions by donating electrons to neutralize reactive oxygen species (ROS).[3] Consequently, the observed decrease in ascorbic acid concentration directly correlates to a reduction in the spinach's ability to support collagen synthesis and provide cellular antioxidant protection. The interval of 2 minutes between each change was chosen because of it's real-world significance: 0 minutes is a raw baseline, 2 minutes reflects the realistic time for blanching, 4 minutes reflects standard cooking time, 6 minutes reflects stewing, and 8 minutes reflects pureeing. Anything past 8 minutes simply reduces the spinach into unusable mush.

To measure the ascorbic acid concentration in spinach, this experiment uses an iodine titration method. Ascorbic acid reacts with iodine in a redox reaction, allowing its concentration to be determined quantitatively from the titre volume. This method is suitable for the investigation because it is precise, accessible in a school laboratory, and effective for analyzing ascorbic acid in food extracts.

1.1 Research Question

To what extent does the cooking time influence the ascorbic acid content in spinach, as quantified by a redox titration?

1.2 Hypothesis

It is hypothesized that increasing the boiling time of spinach from 0 to 8 minutes will reduce the concentration of ascorbic acid remaining in the spinach aliquot. Ascorbic acid is susceptible to oxidation and thermal decomposition, and because it is highly water-soluble, prolonged boiling also promotes its transfer from the spinach tissue into the cooking water. Consequently, greater cooking times should result in progressively lower vitamin C concentrations.

1.3 Safety Considerations

To ensure safety, goggles and lab coats were worn to protect against 0.5 M and 6 M HCl, and 0.001 M iodine and iodine crystals. All HCl handling was conducted in an enclosed space and under a fume hood, while beaker tongs were utilized to prevent thermal burns from $100°C$ boiling water. Environmentally, liquid titration residues were collected in hazardous waste containers, while solid spinach pulp was disposed of as organic waste. Any accidental chemical contact was immediately flushed with water for 15 minutes.

2 Variables

2.1 Independent Variable

The boiling time of the spinach (in minutes) increased by an interval of 2 minutes and starting at 0. (0, 2, 4, 6, 8)

2.2 Dependent Variable

The concentration of ascorbic acid (in $\mathrm{mol\,dm^{-3}}/M$).

2.3 Controlled Variables

Table 1: Controlled variables

Controlled variable	**Why it needs to be controlled**	**How it is controlled**
Boiling temperature	Temperature affects the rate of thermal degradation and oxidation of ascorbic acid. Therefore, higher temperatures increase reaction rates, which would alter the measured ascorbic acid concentration independently of boiling time.	To control this variable, the same hot plate setting (Medium - 5) was used for all trials, maintaining a consistent rolling boil throughout each experiment and placing the spinach into the water when the thermometer read an even 100°C

Controlled variable	**Why it needs to be controlled**	**How it is controlled**
Mass of spinach	The total amount of ascorbic acid present is directly proportional to the mass of spinach used. Varying the mass would change the concentration measured, making comparisons between trials invalid.	A constant mass of 5.00 ± 0.01 g (While the 3-decimal digital balance had an uncertainty of ± 0.001 g, a higher uncertainty of ± 0.01 g was assigned to account for the variations during weighing likely caused by air currents in the lab and the loss due to evaporation of the fresh spinach samples.) of spinach was measured using a digital balance for each trial.
Volume of boiling water	The volume of water affects dilution and the extent of ascorbic acid leaching from the spinach into the surrounding liquid. Different volumes would alter the final concentration of ascorbic acid.	A fixed volume of 150.0 ± 2mL of distilled water was measured using a 250 mL graduated cylinder for all trials.
Final aliquot volume	Since $$n = cv$$, the calculated concentration depends on the total volume of aliquot.	The total aliquot volume was kept consistent 10.0 ± 0.1 mL and measured using a 10 mL graduated cylinder.
Blending time	Blending affects the efficiency of the ascorbic acid extraction, and inconsistent blending would lead to unequal extraction and unreliable concentrations.	All samples were blended for a constant duration of 1 minute using the same handheld blender.
Filtration method	Different filtration methods can affect the amount of solids filtered out as pulp, ultimately affecting the concentration of the filtrate analyzed.	The same type and brand of 140x140mm filter paper was used for all trials, with consistent filtration time of 30 minutes.
Iodine concentration for titration	The concentration of the iodine solution is the most crucial part to solve for the ascorbic acid content since it is used directly to calculate the concentration of ascorbic acid. Any variation would introduce systematic error in calculations.	A fixed concentration of 0.00100 ± 0.00001 mol dm^{-3} of iodine solution was used for each titration.

Controlled variable	Why it needs to be controlled	How it is controlled
Ascorbic acid leaching	Ascorbic acid is highly water-soluble, so removing/straining the boiling water could alter how much ascorbic acid is calculated since some of the acid can transfer into the surrounding liquid.	In every trial, the spinach and boiling water were processed together and blended before filtration to ensure all leached ascorbic acid was included in the analysis.

3 Methodology

3.1 Materials and Apparatus

Table 2: **M**aterials and Apparatus

Materials	**Apparatus**
Baby Spinach (*Spinacia oleracea*)	250 mL Beaker (1)
0.001 M Iodine solution (I_2)	50 mL Beakers (5)
0.5 M Hydrochloric acid (HCl)	125 mL Erlenmeyer flask (1)
Starch Solute	50 mL Burette (1)
Distilled Water	Burette stand (1) and clamp (1)
140x140mm Filter Paper	250 mL Graduated cylinder (1)
	100 mL Graduated cylinder (1)
	10 mL Graduated cylinder (1)
	10.00 mL Volumetric pipette (1)
	Transfer pipettes (2)
	Blender (1)
	Hot plate (1)
	Thermometer (1)
	Stopwatch (1)
	Plastic Funnel (1)
	3-decimal place electronic balance (1)
	Beaker Tongs (1)
	Glass stirring rod (1)

3.2 Procedure

Preparation of the Starch Indicator Solution

1. Weigh 1.000±0.001g of soluble starch with a piece of brown paper using a scoopula and a 3-decimal electronic balance.

2. Transfer the starch to a 150mL beaker, measure 10.0 ± 0.1mL of cold distilled water using a 10 mL graduated cylinder, and mix the water thoroughly with the starch using a glass stirring rod until a smooth paste is formed.

3. In a separate 250 mL beaker, measure out 90 ± 1mL using a 100 mL graduated cylinder and add to the beaker, bring the distilled water to a rolling boil on a hot plate.

4. Gradually pour the cold starch paste into the boiling water while stirring continuously.

5. Allow the mixture to boil for approximately 1 minute until the solution becomes relatively clear.

6. Remove the beaker from the heat and allow the indicator solution to cool completely to room temperature before using.

Calculation and Preparation of 0.001M Iodine Solution

The experimental concentration of the iodine solution is calculated using the measured 0.127 ± 0.001 g of iodine crystals and the molar mass of diatomic iodine ($M(\mathrm{I_2}) = 253.80\ \mathrm{g\ mol^{-1}}$). The number of moles ($n$) is calculated as:

$$n_{\mathrm{I_2}} = \frac{m}{M} = \frac{0.127\ \mathrm{g}}{253.80\ \mathrm{g\ mol^{-1}}} = 0.000500\ \mathrm{mol}$$

Since the molar mass is treated as an exact theoretical value, the uncertainty in the number of moles is solely based on the measured mass uncertainty:

$$\%\ \text{uncertainty (mass)} = \left(\frac{0.001}{0.127}\right) \times 100 \approx 0.787\%$$

Absolute uncertainty (n) $= 0.000500 \times 0.787\% = \pm 0.0000039\ \mathrm{mol} \approx 0.000004\mathrm{mol}$

Therefore, the moles of iodine are $n_{\mathrm{I_2}} = 0.000500 \pm 0.000004$ mol.

The concentration (c) of the aqueous iodine solution is then calculated using the formula $c = \frac{n}{V}$:

$$c_{\mathrm{I_2}} = \frac{0.000500\ \mathrm{mol}}{0.5000\ \mathrm{dm^3}} = 0.00100\mathrm{M}$$

To find the total uncertainty of the concentration, the percentage uncertainties of the volume and the moles are summed together:

$$\%\ \text{uncertainty (volume)} = \left(\frac{0.0025}{0.5000}\right) \times 100 = 0.500\%$$

$$\%\ \text{uncertainty (concentration)} = 0.787\% + 0.500\% = 1.287\%$$

$$\text{Absolute uncertainty } (c) = 0.00100 \times \left(\frac{1.287}{100}\right) = 0.00001287\mathrm{M}$$

To align the decimal places of the calculated concentration with standard significant figure rules, the uncertainty is rounded to one significant figure. The final experimental concentration of the iodine solution is reported as:

$$c_{\mathrm{I_2}} = 0.00100 \pm 0.00001\mathrm{M}$$

1. Weigh 0.127 ± 0.001 g of iodine crystals on a piece of brown paper using a scoopula and a 3-decimal electronic balance.

2. Transfer the iodine crystals into a 500 mL beaker. Measure 500.0 ± 2.5 mL of distilled water using a 500 mL graduated cylinder and pour it into the beaker with the iodine.

3. Stir the mixture vigorously using a glass stirring rod to begin dissolving the crystals. Continue to stir the solution continuously until all solid iodine crystals have completely dissolved and the solution reaches a uniform yellow-brown color.

4. Store it in a dark space until the titrations begin, as iodine solutions can degrade when exposed to light.

Preparation of 0.5 M Hydrochloric Acid

To prepare the 0.5M hydrochloric acid (HCl) required to stabilize the ascorbic acid for the titration, a stock solution of $6.0M$ HCl was diluted.

1. Measure 10.0 ± 0.1mL of the 6.0 M HCl solution in a graduated cylinder and transfer to a 250 mL beaker.
2. Measure 110 ± 2mL using a 250 mL graduated cylinder and transfer into the HCl beaker

The final volume of 120 mL was made to satisfy all the trials in the experiment for a week to avoid heavy degradation (repeated once more). The final concentration was calculated using the dilution equation, $C_1V_1 = C_2V_2$, rearranged to solve for the final volume (C_2):

$$C_2 = \frac{C_1V_1}{V_2} \tag{1}$$

$$C_2 = \frac{(6.0\text{M})(0.0100\ \text{dm}^3)}{0.120\ \text{dm}^3} = 0.50\text{M}$$

$$\%\ \text{uncertainty}\ (V_1) = \left(\frac{0.0002}{0.0100}\right) \times 100 = 2.00\%$$

$$\%\ \text{uncertainty}\ (V_2) = \left(\frac{0.001}{0.120}\right) \times 100 = 0.83\%$$

$$\%\ \text{uncertainty}\ (C_2) = 2.00\% + 0.83\% = 2.83\%$$

The absolute uncertainty of the diluted HCl concentration is then calculated as:

$$\text{Absolute uncertainty}\ (C_2) = 0.50 \times \left(\frac{2.83}{100}\right) = 0.01415\text{M}$$

Rounding the uncertainty to one significant figure (± 0.02) to align decimal places, the final calculated concentration for the diluted hydrochloric acid is: $\boxed{C_{\text{HCl}} = 0.50 \pm 0.01\text{M}}$

Setup

1. Keep the following variables constant for every trial:
   - mass of spinach: 5.00 g
   - boiling water volume: 150.0 mL
   - filtrate aliquot volume per titration: 10.00 mL
   - iodine concentration: $0.001M$
   - hydrochloric acid added before titration: 5.0 mL of $0.5M$ HCl
2. Set up a hot plate at the 5th heating interval (Medium heat or roughly 350°F to 400°F).
3. Using the 250 mL graduated cylinder measure 150 ± 2mL of distilled water and pour into a 250 mL beaker.
4. Bring the water inside the beaker to a boil. Verify that the water has reached a temperature of 100 °C using a thermometer.
5. Place the 5.00 ± 0.01g of baby spinach into the boiling water and swirl using a glass stirring rod and start the stopwatch immediately when the spinach makes contact with the water.

6. Remove the beaker using the beaker tongs immediately from the heat place once the stopwatch has reached:
    - 0 min
    - 2 min
    - 4 min
    - 6 min
    - 8 min
7. Leave the beaker to cool on the table for 10 minutes.
8. Pour the liquid mixture (water + spinach) into the blender and blend at high settings for 1 minute.
9. Afterwards, pour the mixture liquid on filter paper on top of a 250 mL beaker.
10. Wait 30 minutes for the filter paper to separate the solids and the liquids then separate the liquid using a pipette, a 10 mL graduated cylinder and 5 50 mL beakers. Measure 10 mL of spinach aliquot, using the pipette and the 10mL graduated cylinder.

Ascorbic Acid Titration

1. Pipette 10.00 mL of spinach filtrate into a conical flask labelled for ascorbic acid analysis.
2. Add in:
    - 20 mL distilled water
    - 5.0 mL dilute HCl
    - 3-4 drops of starch indicator
3. Titrate the solution with 0.001 mol dm^{-3} iodine solution until the mixture becomes pale yellow.
4. Continue adding iodine dropwise until the first permanent blue-black colour persists for 10–15 s.
5. Record the titre volume as V_I.
6. Repeat Setup Step 5 to Titration Step 5 until all 5 trials are completed.

4 Data Collection

After repeating the experiment 5 times for each of the 5 intervals, every measurement and uncertainty was measured and placed into an excel file. The iodine titration volume was found by marking the initial burette volume and subtracting the final burette volume, accounting for uncertainty.

4.1 Average Titre Calculation

This sample calculation will determine the average titre volume of the aliquot of spinach boiled for 2 minutes.

1. Find the titre value for each of the five trials.

$$\text{titre value} = \text{final burette volume} - \text{initial burette volume}$$
$$\text{titre value} = 0.0270\,\text{dm}^3 - 0.0241\,\text{dm}^3$$
$$\text{titre value} = 0.0029\,\text{dm}^3$$

$$\text{uncertainty} = \pm 0.00005\,\text{dm}^3 + \pm 0.00005\,\text{dm}^3 \text{uncertainty} \quad = \pm 0.0001\,\text{dm}^3$$

$$\boxed{\text{titre value} = 0.0029 \pm 0.0001\,\text{dm}^3}$$

2. Using the five titre volumes, the average titre volume was determined using the equation:

$$\bar{X} = \frac{\sum X}{n} \tag{2}$$

(Note: The n in this equation does not represent the amount of moles, but rather the number of trials.)

$$\bar{X} = \frac{\sum X}{n}$$
$$\bar{X} = \frac{0.0029\,\text{dm}^3 + 0.0036\,\text{dm}^3 + 0.0026\,\text{dm}^3 + 0.0030\,\text{dm}^3 + 0.0025\,\text{dm}^3}{5 \text{ trials}}$$
$$\bar{X} = 0.00292\,\text{dm}^3$$

3. The uncertainty for the average titre volume was determined in the same manner as the volumes:

$$\Delta\bar{X} = \frac{\sum \Delta X}{n} \tag{3}$$

(Note: the n in this equation does not represent the amount in moles, but rather the number of trials.)

$$\Delta\bar{X} = \frac{\sum \Delta X}{n}$$
$$\Delta\bar{X} = \frac{0.0001\,\text{dm}^3 + 0.0001\,\text{dm}^3 + 0.0001\,\text{dm}^3 + 0.0001\,\text{dm}^3 + 0.0001\,\text{dm}^3}{5 \text{ trials}}$$
$$\Delta\bar{X} = 0.0001\,\text{dm}^3$$

$$\text{titre value} = 0.0029 \pm 0.0001\,\text{dm}^3$$

(value rounded to match the number of decimal places as uncertainty)

4. The standard deviation of each change and its 5 trials: An example calculation will be made using the 2-minute time interval, whereas the rest of the standard deviations will be calculated separately and placed on the table. The standard deviation (s) was calculated using the following formula:

$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}} \tag{4}$$

Where:

- x_i represents each individual titre volume
- $\bar{x} = 0.00292$ dm^3 (the average titre volume)
- $n = 5$ (the number of trials)

First, the sum of the squared differences from the mean is calculated:

$$\begin{aligned}\sum(x_i - \bar{x})^2 = (0.0029 - 0.00292)^2 + (0.0036 - 0.00292)^2 \\ + (0.0026 - 0.00292)^2 + (0.0030 - 0.00292)^2 \\ + (0.0025 - 0.00292)^2\end{aligned}$$

$$\sum(x_i - \bar{x})^2 = 7.48 \times 10^{-7}\ \text{dm}^6$$

This sum is then substituted back into the sample standard deviation formula:

$$s = \sqrt{\frac{7.48 \times 10^{-7}\ \text{dm}^6}{5-1}}$$

$$s = \sqrt{1.87 \times 10^{-7}\ \text{dm}^6} = 0.0004324\ldots\ \text{dm}^3 \approx 0.00043\ \text{dm}^3$$

Table 3: Processed iodine titration data for each spinach boiling time

Time Cooked	**Mean Mass of Spinach** ± 0.01 **(g)**	**Mean Volume After Blending** ± 2.5 ($\times 10^{-3}$dm^3)	**Mean Iodine Titration** ± 0.1 ($\times 10^{-3}$dm^3)	**Standard Deviation** ($\times 10^{-3}$dm^3)
No Time	5.00	145	4.60	0.82
2 min	5.00	145	2.92	0.43
4 min	5.00	145	1.90	0.19
6 min	5.00	145	1.84	0.29
8 min	5.00	145	1.46	0.15

I. Processed Data Interpretation

From the provided data in Table 3, there is a strong negative relationship between the length of boiling time and the average titre volume of iodine required to reach the equivalence point. The data demonstrates that the greatest average titre value (4.60 mL) was required for the raw, unboiled spinach aliquot, indicating the highest initial concentration of ascorbic acid in baby spinach. As boiling times progressively increased, the required titre volumes steadily decreased. The least average titre value (1.46 mL) was required for the aliquot of spinach boiled for 8 minutes. This downward trend confirms the hypothesis that extended thermal exposure degrades the ascorbic acid in spinach tissues.

II. Qualitative Observations

- **Color Interference:** Due to the natural chlorophyll pigmentation in spinach, the starch indicator was not as visually clear when titrating. Therefore, the spinach mixture was diluted with a 1:1 ratio of distilled water. This effectively reduced the intense green colour to a clear green. Because of this dilution step, the sudden appearance of the deep blue-black starch indicator at the equivalence point was highly visible against the pale background, significantly improving the precision of the final burette readings.
- **Approaching the Endpoint:** As the solution approached the equivalence point and the ascorbic acid concentration depleted, the rate of color dissipation significantly decreased. The flask required much more vigorous mixing for a longer duration to return to its pale-green state. The final endpoint was marked by a distinct, permanent color shift where a dark blue-black hue completely overtook the solution and persisted for at least 10–15 seconds, indicating that all ascorbic acid had been oxidized and the excess iodine had successfully reacted with the starch.

4.2 Stoichiometry: The Concentration of Ascorbic Acid

The qualitative observations and data were calculated using solely the measurements from the Iodine Titration. This data is then used to calculate the concentration of ascorbic acid. The chemical equation is:

$$\mathrm{C_6H_8O_6 + I_2 -> C_6H_6O_6 + 2\,H^+ + 2\,I^-} \tag{5}$$

From the titration equation, a clear 1:1 ratio between the ascorbic acid and the iodine can be observed. This means for 1 mol of iodine used to titrate, 1 mol of ascorbic acid reacted with it.

1. Using the titre volume, calculate the number of moles $n = cv$ In each of the experiments, the volume of each aliquot was 10 ± 0.5mL. Therefore by using the mol value from the earlier section and the standard volume of each aliquot, the concentration of ascorbic acid in each trial can be calculated. An example calculation for both the concentration of one trial and the mean concentration will be done with the 2 minute interval. The number of moles of iodine (n_I) is calculated using the average titre volume and the known concentration:

$$n_I = v_I \times c_I$$

$$n_I = (0.00292\ \mathrm{dm^3}) \times (0.001\mathrm{M})$$

$$n_I = 2.92 \times 10^{-6}\ \mathrm{mol}$$

Using the 1:1 stoichiometric ratio, the concentration of ascorbic acid in the 10 mL ($0.010\ \mathrm{dm^3}$) sample is:

$$c_A = \frac{n_I}{v_A}$$

$$c_A = \frac{2.92 \times 10^{-6}\ \mathrm{mol}}{0.0100\ \mathrm{dm^3}}$$

$$c_A = 2.92 \times 10^{-4}\mathrm{M}$$

The total percentage uncertainty of c_A is the sum of the individual percentage uncertainties of these three variables.

$$\% \text{ uncertainty } (v_I) = \left(\frac{0.0001}{0.00292}\right) \times 100 = 3.42\%$$

$$\% \text{ uncertainty } (c_I) = \left(\frac{0.00001}{0.00100}\right) \times 100 = 1.00\%$$

$$\% \text{ uncertainty } (v_A) = \left(\frac{0.0001}{0.0100}\right) \times 100 = 1.00\%$$

Next, these percentage uncertainties are summed:

$$\% \text{ uncertainty } (c_A) = 3.42\% + 1.00\% + 1.00\% = 5.42\%$$

Using the corrected calculated concentration of $c_A = 2.92 \times 10^{-4}$ mol dm^{-3}, the absolute uncertainty is determined:

$$\text{Absolute uncertainty } (c_A) = (2.92 \times 10^{-4}) \times \left(\frac{5.42}{100}\right) = 1.58264 \times 10^{-5}\text{M}$$

Rounding the absolute uncertainty to one significant figure ($\pm 2 \times 10^{-5}$) to represent the precision limit, the final experimental concentration of ascorbic acid:

$$c_A = (2.9 \pm 0.2) \times 10^{-4}\text{M}$$

2. Ascorbic acid concentration table

Table 4: Ascorbic Acid Concentration

Time Cooked	**Concentration of Ascorbic Acid ($\times 10^{-4}$ M)**
No Time	4.6 ± 0.4
2 min	2.9 ± 0.2
4 min	1.9 ± 0.2
6 min	1.8 ± 0.2
8 min	1.5 ± 0.2

5 Data Processing

The data shows that the concentration of ascorbic acid decreases as the time boiled increases, however at what rate does it degrade? To determine the most accurate reaction order for the thermal degradation of ascorbic acid, the experimental data was fitted against zero, first, and second-order integrated rate laws. The goal was to identify the transformation of ascorbic acid concentration that yielded the highest coefficient of determination (R^2). The zero-order model assumes the rate of degradation is independent of concentration; a linear relationship. This is represented by:

$$[A]_t = -kt + [A]_0 \tag{6}$$

A plot of $[AA]$ versus time was generated and while showing a clear downward trend, the R^2 value ≈ 0.91 suggests that the rate is not constant, as the degradation slows significantly at lower concentrations. The first-order model assumes the rate is proportional to the remaining concentration of ascorbic acid. The integrated rate law is:

$$\ln[A]_t = -kt + \ln[A]_0 \tag{7}$$

By plotting ln[A] versus time, a more linear relationship was observed: $R^2 \approx 0.95$. This suggests that the loss of ascorbic acid is a first-order process, likely due to the molecules being consumed by oxidation or diffusing into the solvent at a rate proportional to their availability in the plant tissue. A second-order model assumes the rate is proportional to the square of the concentration:

$$\frac{1}{[A]_t} = kt + \frac{1}{[A]_0} \tag{8}$$

The model was observed to have a $R^2 \approx 0.92$, therefore, the first-order model was selected as the most realistic and statistical model for this experiment. The plot of ln[A] versus time produced a more linear relationship r = -0.9573 indicating that the data follows apparent first-order kinetics. The results suggest that the rate of vitamin C loss is proportional to its concentration. This observation reflects the collision theory, where a reduced probability of a collision with enough energy resulted in a reaction. In this case, the oxidation of ascorbic acid is proportional to the concentration of ascorbic acid, because of the probability that molecules collide and react with one another within the beaker during the boiling process as time progresses. This concludes that it is much more beneficial nutritionally to eat raw spinach, then boil it whatsoever.

6 Evaluation

6.1 Strengths

One of the key strengths of this experiment was the ability to conclusively observe a negative linear relationship between the concentration of ascorbic acid and the boiling time. This was proven through multiple trials, with consideration for the uncertainty. In addition, the rate-order observation helped combine theory with realistic lab observations and helped create the collision theory connection. The last key strength was the minimization of human errors through experimentation, mainly, the use of dilution to reduce the impact of sight human-error.

6.2 Limitations

Unfortunately, the experiment ran into several limitations as a result of the equipment and lab setting. A significant limitation that occurred was as a result of the oxidation of spinach, even in the closed-refrigerated container. Likewise, the iodine solution was slowly degraded over time in the lab, even if the solution was kept in a dark and stopper-covered beaker. More limitations included the filtration step, since the filter paper only extracted the liquid, the pulp could have contained some vitamin C. The other limitations included the titration method since polyphenols and flavonoids can also reduce iodine, the iodine titration did not only react with ascorbic acid, instead it reacted with all of these compounds. Lastly, the last limitation was an uncontrollable case of human error during the titration, as the starch endpoint is visual, the endpoint of the titration is entirely judgment-based.

6.3 Improvements

Improvements to the lab include using more accurate equipment, such as a High-Performance Liquid Chromatography (HPLC). This is a more accurate method which correctly identifies the ascorbic acid content separately from the flavonoids and polyphenols. A second major benefit of this method is the elimination of sight human-error, since the data is calculated rather than observed. This improvement would massively

improve the accuracy of the experiment and eliminate the biggest source of human error in the titration method. Lastly, the oxidation of the spinach during transportation and the experiment could have been reduced using 1,4-dithiothreitol (DTT). However, due to the specialized nature and limited availability of industrial reducing agents like DTT, a total ascorbic acid analysis could not be performed. The investigation was therefore restricted to measuring the concentration of non-oxidized ascorbic acid.

References

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